

CBT Exam Hall — Network Setup Runbook

Router: TP-Link **TL-WR840N** (SSID currently **TP-Link_9552**)
Server: Windows Server 2022 (Hyper-V host) + Ubuntu VM running the CBT stack
Clients: Wireless exam workstations

0. Hardware profile — read before configuring

TP-Link_9552 is the **factory default SSID** — TP-Link names it **TP-Link_XXXX** using the last four hex digits of the router's MAC. The router has never been properly configured, which explains the addressing confusion.

Check the sticker underneath for the **hardware revision** (V1 through V6+). Menu wording shifts slightly between revisions; the paths below match the classic green TP-Link web UI used across this range.

Item	Value
Model	TL-WR840N — N300, single-band
Radio	2.4 GHz only — there is no 5 GHz radio
Wireless standard	802.11b/g/n, up to 300 Mbps (theoretical)
Ethernet	1 × WAN + 4 × LAN, all 10/100 Mbps — not Gigabit
CPU	Single-core, entry-level
Max encryption	WPA2-PSK (no WPA3)
Router admin URL	http://192.168.0.1 or http://tplinkwifi.net
Default login	admin / admin
Subnet	192.168.0.0/24 , mask 255.255.255.0

Three constraints this model imposes

- 1. Single 2.4 GHz band.** Every client competes for the same spectrum. There is no 5 GHz to offload capable machines onto, so the usable client count is lower than a dual-band router and interference from phones, microwaves, and neighbouring WiFi hits everything at once.
- 2. 100 Mbps ceiling to the server.** The server's ethernet link runs at 100 Mbps, and that is the *total* pipe shared by every candidate. For text-based papers this is fine. For image-heavy questions it isn't — 30 clients each pulling a 2 MB page simultaneously takes roughly 5 seconds just in transfer. Mitigations: compress and resize all question images, enable gzip/brotli on the app, and stagger candidate logins rather than starting everyone at once.
- 3. Realistic client capacity is around 15-25 machines.** This is a budget home router with a single-core CPU; it was designed for a small flat. Push it past that and you get association failures and timeouts, not graceful slowdown.

If the hall exceeds ~20 machines, buy a second TL-WR840N (or better), set it to **Access Point mode**, and connect it to a LAN port of the first with an ethernet cable. Give it the same SSID and password but a different channel (1 and 11, say). Wired backhaul is essential — do not use WDS/repeater mode, which halves throughput. Note you only have 4 LAN ports total, one of which goes to the server.

Run a full-load rehearsal. Everything below can be configured perfectly and still fail at 25 clients because of the hardware.

1. Address plan

Fix this on paper before touching anything. Every later step refers back to it.

Range	Assignment	Method
192.168.0.1	Router / gateway	Router LAN config
192.168.0.2	Spare (second AP / switch)	Static
192.168.0.3	Windows Server 2022 host — management only	Static
192.168.0.4	Ubuntu VM — CBT services (students connect here)	Static
192.168.0.5 — .19	Printers, infrastructure	Static
192.168.0.20 — .99	Reserved / future static	—
192.168.0.100 — .200	Exam client pool	DHCP
192.168.0.201 — .254	Unused headroom	—

Critical point: students point their browsers at the **VM** (192.168.0.4), not the Windows host. The host only runs Hyper-V. Getting these two confused is the single most common failure in this architecture.

2. Router configuration

Log in at <http://192.168.0.1>.

On this model the left-hand menu is a flat tree: *Status, Quick Setup, WPS, Network, Wireless, DHCP, Forwarding, Security, Parental Controls, Access Control, Advanced Routing, Bandwidth Control, IP & MAC Binding, Dynamic DNS, System Tools*. Note that **DHCP is its own top-level item**, not nested under Network.

2.1 Secure the router

- **System Tools → Password** (older revisions) or **System Tools → Administration** — set a strong admin password. The factory `admin / admin` must not survive. Record it on the handover sheet.
- **System Tools → Firmware Upgrade** — check your revision on tp-link.com and update **now**, never within a week of an exam. Download the firmware matching your exact hardware revision; flashing the wrong one bricks the unit.
- **WPS → Disable WPS** (top-level menu on this model). It's a standing security hole and serves no purpose here.
- **Guest Network → Disable**.

2.2 LAN and DHCP

Network → LAN

- IP Address: 192.168.0.1
- Subnet Mask: 255.255.255.0

DHCP → DHCP Settings (top-level DHCP menu)

Field	Value
DHCP Server	Enable
Start IP	192.168.0.100
End IP	192.168.0.200
Lease Time	1440 minutes (24 hours)
Default Gateway	192.168.0.1
Primary DNS	192.168.0.1

Saving LAN changes reboots the router. Reconnect afterwards.

2.3 Wireless

Wireless → Basic Settings (called *Wireless Settings* on some revisions)

Setting	Value	Why
Wireless Network Name	CBT-HALL-A	Identifiable; stop broadcasting factory defaults
Region	Nigeria	Sets legal channel and power limits
Mode	11bgn mixed	Drop to 11bg only if very old clients fail to associate
Channel	Fixed: 1, 6, or 11 — never Auto	Auto re-scans and drops clients mid-exam
Channel Width	20 MHz	40 MHz on 2.4 GHz consumes two of only three clean channels and is unstable under load
Enable SSID Broadcast	Checked	Hiding it causes more support calls than it prevents

Scan the hall first with a WiFi analyser app on a phone and pick whichever of 1, 6, or 11 is least occupied. Those are the only three non-overlapping channels in 2.4 GHz — never choose 3, 4, 8, or anything between.

Wireless → Wireless Security

Setting	Value
Mode	WPA2-PSK
Version	WPA2-PSK
Encryption	AES (never TKIP — it caps the link at 54 Mbps)
Password	12+ characters, no spaces

WPA2-PSK is the strongest option this model offers; there is no WPA3.

2.4 The setting that caused your ping problem

Wireless → Wireless Advanced → Enable AP Isolation — must be **unchecked**.

AP Isolation prevents wireless clients from reaching each other and, on some firmware, from reaching wired hosts. It produces exactly the symptom you started with. Not every TL-WR840N revision exposes this option — if you don't see it on the Wireless Advanced page, your firmware doesn't implement it and there's nothing to change.

Also on this page, leave **WMM enabled** — disabling it degrades throughput on 802.11n.

2.5 Address reservations — not needed for the server

Skip reservations for 192.168.0.3 and 192.168.0.4. Most TL-WR840N firmware silently ignores a reservation pointing outside the DHCP pool, so reserving .3 against a .100 - .200 pool does nothing. More importantly, it's unnecessary: those two machines are configured statically on the machines themselves (sections 3 and 4) and never request an address from the router at all.

If a machine you expected to be static is showing up with a 192.168.0.1xx address, it is still on DHCP — the static configuration wasn't applied, or was applied to the wrong adapter. Fix it on the machine, not on the router.

Because the static addresses sit at .3 and .4 while the pool starts at .100, the ranges cannot collide. No reservation is required to keep them apart.

DHCP → DHCP Client List is still useful — it shows every connected device and its MAC, which is the fastest way to confirm what the router is actually handing out.

2.6 Internet uplink — a decision to make

Leaving the WAN port unplugged gives you a fully isolated exam network: no browsing, no updates mid-exam, no bandwidth contention. The trade-off is that you cannot pull Docker images or reach external services while offline.

Recommended: keep WAN connected during setup and image pulls, unplug it before the exam begins. Confirm beforehand that the CBT stack has no external dependencies at runtime (fonts, CDNs, analytics, licence checks).

3. Windows Server 2022 host

3.1 Verify the Hyper-V virtual switch

```
Get-VMSwitch
```

You need one external switch bound to the physical NIC, sharing it with the host:

```
# Only if it doesn't already exist
New-VMSwitch -Name "ExternalSwitch" -NetAdapterName "Ethernet" -AllowManagementOS $true
```

3.2 Assign the host IP to the *virtual* adapter

Once an external vSwitch exists, the host's IP lives on `vEthernet (ExternalSwitch)` — **not** on the physical NIC. Putting it on the physical adapter is the classic cause of "it's configured but nothing works."

```
# Clear any stale configuration first
Remove-NetIPAddress -InterfaceAlias "vEthernet (ExternalSwitch)" -Confirm:$false -ErrorAction SilentlyContinue
Remove-NetRoute -InterfaceAlias "vEthernet (ExternalSwitch)" -DestinationPrefix 0.0.0.0/0 -Confirm:$false -ErrorAction SilentlyContinue

# Apply
New-NetIPAddress -InterfaceAlias "vEthernet (ExternalSwitch)" `
  -IPAddress 192.168.0.3 -PrefixLength 24 -DefaultGateway 192.168.0.1

Set-DnsClientServerAddress -InterfaceAlias "vEthernet (ExternalSwitch)" `
  -ServerAddresses 192.168.0.1,8.8.8.8
```

3.3 Confirm exactly one adapter holds an address

```
Get-NetIPConfiguration | Where-Object { $_.IPv4Address }
```

Expected: one entry, `vEthernet (ExternalSwitch)`, `192.168.0.3`, gateway `192.168.0.1`. Any stray `169.254.x.x` or second `192.168.0.x` will cause asymmetric routing — disable that adapter.

3.4 Set the network profile

```
Get-NetConnectionProfile
Set-NetConnectionProfile -InterfaceAlias "vEthernet (ExternalSwitch)" -NetworkCategory Private
Enable-NetFirewallRule -Name FPS-ICMP4-ERQ-In
```

4. Ubuntu VM (CBT services)

4.1 Confirm the VM is on the external switch

```
Get-VMNetworkAdapter -VMName "<YourVMName>"
```

`SwitchName` must read `ExternalSwitch`. If not:

```
Connect-VMNetworkAdapter -VMName "<YourVMName>" -SwitchName "ExternalSwitch"
```

4.2 Static IP via netplan

Find the interface name:

```
ip -br a
```

Edit the netplan file (typically `/etc/netplan/00-installer-config.yaml` or `50-cloud-init.yaml`):

```
network:
  version: 2
  ethernets:
    eth0:
      # replace with your actual interface name
      dhcp4: no
      addresses:
        - 192.168.0.4/24
      routes:
        - to: default
          via: 192.168.0.1
      nameservers:
        addresses: [192.168.0.1, 8.8.8.8]
```

Apply and verify:

```
sudo chmod 600 /etc/netplan/*.yaml
sudo netplan apply
ip -br a
ping -c 3 192.168.0.1
```

4.3 Firewall

Traffic to the VM does **not** pass through the Windows host firewall — an external vSwitch bridges it directly. `ufw` on the VM is what governs access.

```
sudo ufw allow from 192.168.0.0/24 to any port 22 proto tcp
sudo ufw allow from 192.168.0.0/24 to any port 3000 proto tcp # student frontend
sudo ufw allow from 192.168.0.0/24 to any port 7000 proto tcp # admin dashboard
sudo ufw allow from 192.168.0.0/24 to any port 8000 proto tcp # backend API
sudo ufw enable
sudo ufw status verbose
```

Adjust the ports to match your `docker-compose.yml`.

4.4 Docker port binding

In `docker-compose.yml`, published ports must bind to all interfaces:

```
ports:
  - "3000:3000"          # correct – reachable from the LAN
  # - "127.0.0.1:3000:3000" # wrong – loopback only
```

Confirm after `docker compose up -d`:

```
docker compose ps
sudo ss -tlnp | grep -E '3000|7000|8000'
```

You want `0.0.0.0:3000`, not `127.0.0.1:3000`.

5. Client workstations

Leave every exam client on DHCP. Nothing is configured per-machine except the network profile.

```
Set-NetConnectionProfile -NetworkCategory Private
Enable-NetFirewallRule -Name FPS-ICMP4-ERQ-In # optional, diagnostics only
```

Then connect to `CBT-HALL-A` and browse to:

```
http://192.168.0.4:3000
```

Set that as the browser homepage and, if you use kiosk mode, as the launch URL.

6. Verification checklist

Run in order. Stop at the first failure — each step depends on the one above.

#	Check	Command	Expected
1	Host is genuinely static	<code>Get-NetIPInterface -AddressFamily IPv4</code>	Dhcp: Disabled on the live adapter
2	VM has correct IP	<code>ip -br a (VM)</code>	192.168.0.4/24
2	VM reaches gateway	<code>ping -c 3 192.168.0.1 (VM)</code>	Replies
3	Services listening	<code>sudo ss -tlnp \ grep 3000 (VM)</code>	0.0.0.0:3000
4	App responds locally	<code>curl -I http://localhost:3000 (VM)</code>	200 or 3xx
5	Host reaches VM	<code>Test-NetConnection 192.168.0.4 -Port 3000</code>	TcpTestSucceeded: True
6	Client gets a lease	<code>ipconfig /all (client)</code>	192.168.0.100–200
7	Client reaches VM	<code>Test-NetConnection 192.168.0.4 -Port 3000</code>	TcpTestSucceeded: True
8	App loads	Browser → <code>http://192.168.0.4:3000</code>	Login page
9	No IP conflicts	<code>arp -a (host)</code>	No duplicate MACs or IPs
10	Load test	10+ clients logged in simultaneously	No timeouts

Step 5 succeeding while step 7 fails points at the router (AP Isolation, or the client on the wrong SSID). Step 4 succeeding while step 5 fails points at `ufw` or Docker port binding.

7. Handover sheet

Fill this in and leave a printed copy with the on-site IT staff.

Item	Value
Router admin URL	<code>http://192.168.0.1</code>
Router admin password	_____
SSID	CBT-HALL-A
WiFi password	_____
Windows host IP	192.168.0.3
CBT VM IP	192.168.0.4
Student URL	<code>http://coez.cbt.com:3000</code> (or <code>http://192.168.0.4:3000</code>)
Admin dashboard URL	<code>http://coez.cbt.com:7000</code> (or <code>http://192.168.0.4:7000</code>)
Restart services	<code>docker compose restart</code> (on VM)
Escalation contact	_____

8. Exam-day quick fixes

Symptom	Likely cause	Action
One client can't load the app	Wrong SSID, or no lease	<code>ipconfig /release && ipconfig /renew</code>
All clients lost connection	Router reboot / channel change	Check router power; verify channel is fixed, not Auto
App loads but hangs	Docker container down	<code>docker compose ps</code> then <code>restart</code> on the VM
Client got 169.254.x.x	DHCP pool exhausted or router down	Check pool size vs client count
Two machines conflict	Manual IP inside DHCP range	Return the machine to automatic addressing
New clients can't associate	Router at its client limit	Check Wireless → Wireless Statistics for the connected count
Everything slow at peak	100 Mbps uplink saturated	Stagger logins; this is a hardware ceiling, not a setting
Router unresponsive / needs reboot	Single-core CPU overloaded	Power-cycle; reduce client count on this AP

Wireless → Wireless Statistics is your live gauge on exam day — it lists every associated client. If that number is climbing past 20 and complaints are starting, you've found the cause.

Keep one spare workstation configured and idle. It's faster to move a candidate than to debug a failing machine mid-exam.

9. Friendly hostname instead of an IP address

Goal: candidates type `coez.cbt.com:3000` rather than `192.168.0.4:3000`.

Choose the suffix carefully. `cbt.com` is a real registered domain. Locally it resolves correctly because your DNS answers authoritatively for the zone, but any client that falls back to a public resolver will reach the genuine `cbt.com` instead. **`cbt.internal` or `cbt.lan` are reserved for private use** and carry no such risk. Everything below works identically with either — substitute your chosen suffix throughout.

9.1 Method — hosts file entry

No server role, no dependency on anything staying up. Each client holds the mapping itself.

On each workstation, in an **administrator** PowerShell:

```
Add-Content -Path "$env:SystemRoot\System32\drivers\etc\hosts" -Value "`n192.168.0.4`tncoez.cbt.com"
ipconfig /flushdns
```

Verify:

```
Resolve-DnsName coez.cbt.com
```

Why this is the safer choice for exams: if the Windows host fails mid-sitting, hosts-file clients keep resolving the name and the exam continues. With a central DNS server, the host going down takes every candidate offline at once.

Deploying to 20 machines. Put this in a `.ps1` on a USB stick and run it as administrator on each:

```
$hosts = "$env:SystemRoot\System32\drivers\etc\hosts"
$entry = "192.168.0.4`tncoez.cbt.com"
if (-not (Select-String -Path $hosts -Pattern "coez.cbt.com" -Quiet)) {
    Add-Content -Path $hosts -Value "`n$entry"
}
ipconfig /flushdns
Write-Host "Done." -ForegroundColor Green
```

The `if` guard makes it safe to run twice — no duplicate entries.

Limitation: a hosts file maps names to addresses only. It cannot include a port, so candidates still type `coez.cbt.com:3000`. To remove the port too, see 9.3.

9.2 Troubleshooting the hostname

Symptom	Cause	Fix
Name doesn't resolve on one client	Stale DNS cache	<code>ipconfig /flushdns</code>
<code>Access denied</code> writing the hosts file	PowerShell not elevated	Re-open as Administrator
Resolves to a public address	Entry not saved, or typo in the hostname	Open the hosts file in Notepad and check the last line
Name works, page won't load	App or firewall issue, not DNS	Test <code>http://192.168.0.4:3000</code> directly — if that fails too, the problem is elsewhere

The hosts file lives at `C:\Windows\System32\drivers\etc\hosts` and can be opened in Notepad (running as Administrator) if you prefer to check it by eye. Lines starting with `#` are comments.